

6	(a) probability of decay of a nucleus per unit time (allow 1 mark for $A = \lambda N$, with symbols explained)	M1 A1	[2]
(b)	(i) $\lambda = \ln 2 / (28 \times 365 \times 24 \times 3600)$ $= 7.85 \times 10^{-10} \text{ s}^{-1}$	C1 A1	[2]
	(ii) $A = (-)\lambda N$ $N = (6.4 \times 10^9) / (7.85 \times 10^{-10})$ $= 8.15 \times 10^{18}$ mass = $(8.15 \times 10^{18} \times 90) / (6.02 \times 10^{23})$ (e.c.f. for value of N) $= 1.22 \times 10^{-3} \text{ g}$	C1 C1 C1 A1	[4]
	(iii) volume = $(1.22 \times 10^{-3} / 2.54 =) 4.8 \times 10^{-4} \text{ cm}^3$	A1	[1]
(c)	either very small volume of Strontium-90 has high activity or dust can be highly radioactive breathing in dust presents health hazard	B1 B1	[2]

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7 (a) (i) oscillations are damped/amplitude decreases

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